

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination October 2017

PH119/PH127 Pharmaceutical Chemistry-II

Date: 07.10.17, Saturday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

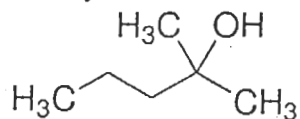
1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

Section - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. Identify the correct IUPAC name for the below structure



- [A] 2 methyl 2 pentanol
[B] 4 methyl 4 pentanol
[C] 2 methyl 2 hexanol
[D] 4 methyl 4 hexanol

2. Saturated hydrocarbons are also referred as

- [A] Alkanes
[B] Alkenes
[C] Alkynes
[D] Alkaloids

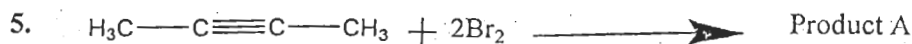
3. Alkyl halides undergo a type of reaction

- P. Nucleophilic substitution
Q. Nucleophilic addition
R. Elimination
S. Electrophilic aromatic substitution

- [A] P and Q are correct
[B] P and R are correct
[C] P, Q, R and S are correct
[D] Q and R are correct

4. Addition of water to alkene will give _____

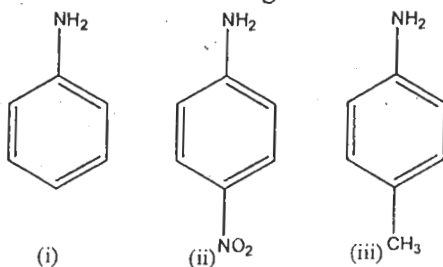
- [A] Aldehyde
[B] Ketone
[C] Alcohol
[D] Carboxylic acid



The product A is..

- [A] 1,2 Dibromo Ethane
- [B] 2,2,3,3 tetra bromo butane
- [C] 2,3 Dibromo butane
- [D] 1,1,2,2 tetra bromo ethane

6. The correct increasing order of basic strength for following compounds is



- [A] ii < iii < i
- [B] iii < ii < i
- [C] i < ii < iii
- [D] ii < i < iii

7. The IUPAC name of isopropyl alcohol is

- [A] 2-Propanol
- [B] 1-Propanol
- [C] 3-Propanol
- [D] 2-Propanal

8. The following statements are correct for carbocation; Except One.

- [A] Stability order of carbocation is $3^\circ > 2^\circ > 1^\circ$
- [B] Electron donating group increase the stability of carbocation
- [C] Carbocation can act as an electrophile
- [D] Electron withdrawing group increase the stability of carbocation

9. Orientation of Elimination reaction follows...

- [A] Markoniov's rule
- [B] Saytzeff rule
- [C] Micheal addition
- [D] Hoffmann rule

10. For which mechanisms the first step involved is the same?

- [A] E_1 and E_2
- [B] E_2 and SN_2
- [C] E_1 and SN_1
- [D] SN_1 and SN_2

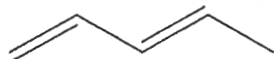
11. Ethylene on reaction with bromine forms which among the following product

- [A] $\text{BrH}_2\text{C}-\text{CH}_2\text{Br}$
- [B] $\text{BrH}_2\text{C}=\text{CH}_2\text{Br}$
- [C] $\text{Br}_2\text{HC}=\text{CHBr}_2$
- [D] $\text{Br}_2\text{HC}-\text{CHBr}_2$

12. Identify the correct sequence according to electronegativity

- [A] $\text{F} > \text{NH}_2 > \text{CH}_3 > \text{OH}$
- [B] $\text{NH}_2 > \text{F} > \text{CH}_3 > \text{OH}$
- [C] $\text{NH}_2 > \text{OH} > \text{CH}_3 > \text{F}$
- [D] $\text{F} > \text{OH} > \text{NH}_2 > \text{CH}_3$

13. What is the hybridization of oxygen in water?
 [A] sp
 [B] sp^2
 [C] sp^3
 [D] sp^3d
14. Which among the following is not a property of Ionic bond?
 [A] Losing of electrons
 [B] Gain of electrons
 [C] Sharing of electrons
 [D] Transfer of electrons
15. Identify the following type's dienes.



- [A] Conjugated Diene
 [B] Isolated Diene
 [C] Cumulated Diene
 [D] None of above
16. All are the examples of Heterogeneous Catalyst. Except One.
 [A] Pd
 [B] Ni
 [C] Pt
 [D] Wilkinson's
17. All statements are correct for SN_2 reaction. Except one
 [A] Follow Second order kinetic
 [B] Rearrangement is possible
 [C] Inversion of configuration takes place
 [D] Single step reaction
18. Following reactive intermediates act as an Electrophile. Except One
 [A] Carbocation
 [B] Nitrene
 [C] Carbene
 [D] Carbanion
19. Identify the types of tautomerism observed in following figure.



- [A] Amine Imine
 [B] Keto Enol
 [C] Lactam Lactim
 [D] Amide Imidic
20. Which of the following is a polar aprotic solvent?
 [A] DMSO
 [B] Ethanol
 [C] Water
 [D] Butanol

SECTION – II

- Q 2** Attempt any TWO of the following;
- A** Define Hybridization. Describe sp^2 hybridization with example. 05
- B** Write in brief about Resonance and Inductive effect. 05
- C** Define following terms; 05
- Sigma (σ) and pi (π) bonds
 - Polar and non-polar bonds
 - Conjugation
 - Tautomerism
 - H- bonding
- Q 3** Attempt any TWO of the following;
- A** Classify different reactive intermediates of carbon and explain stability aspects of carbocation. 05
- B** Compare SN_1 and SN_2 reaction. 05
- C** Construct the structure from the following IUPAC names 05
- 2,3 Dimethyl Butane
 - 2- Butene
 - 1-Propanal
 - Ethanoic acid
 - 2-Chloro butanol

SECTION – III

- Q 4** Attempt any FOUR of the following;
- A** Give any three preparation and reactions of Alkyl halide. 05
- B** Give three preparation and reactions of Alkane 05
- C** Give any three preparation and reactions of Alkenes 05
- D** Explain Markonikov's rule and Peroxide effect in detail. 05
- E** Explain 1, 4 addition Vs. 1, 2 addition in dienes. 05
- F** Write reaction and mechanism for Hoffmann degradation of amide. 05
- Q 5** Attempt any FOUR of the following;
- A** Write reaction, mechanism, orientation and reactivity of Elimination Unimolecular reaction. 05
- B** Explain factor affecting SN_1 and SN_2 reaction. 05
- C** Write any two preparation and reaction of Amine. 05
- D** Explain saytzeff's rule with an examples. 05
- E** Give any two preparation and reaction of Alkyne. 05
- F** Write in detail about the preparation and reaction of Alcohol. 05